

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 9

Evolved Packet Core (EPC)

هسته بسته تکامل یافته (EPC)



Lecture Notes

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In This Module

9.1 Why EPC Was Designed	2
9.2 EPC Architecture Overview	3
9.3 EPC Network Functions	4
9.4 EPC Bearers	11
9.5 Authentication in EPC	14
9.6 Session Management	17
9.7 Mobility Management	18
9.8 Failure Scenarios	21
9.9 Summary	23

9.1 Why EPC Was Designed

The Evolved Packet Core (EPC) was introduced in 3GPP Release 8 as the core network for LTE/4G. Its design was driven by fundamental limitations in legacy core networks (2G/3G circuit-switched + packet-switched domains).

Design Principles

Principle	Legacy Problem	EPC Solution
All-IP Architecture	Separate CS and PS domains, TDM transport	Single flat IP-based core for all services (voice via VoLTE/IMS)
Flat Architecture	Deep hierarchy (RNC, SGSN, GGSN) adding latency	Reduced node count; eNB connects directly to EPC
Control/User Plane Separation	Combined CP/UP in SGSN/GGSN	MME (CP only), S-GW/P-GW (UP with minimal CP)
Multiple RAT Support	Separate cores per RAT	Single EPC serves LTE, WCDMA, CDMA2000, Wi-Fi
Simplified Mobility	Complex handover with many nodes	GTP-based anchoring with fewer hops
Policy & Charging Convergence	Fragmented policy/charging	Integrated PCC architecture (PCRF/PCEF)

Key Benefits

- **Lower latency:** User plane path is shorter (eNB → S-GW → P-GW → Internet)
- **Higher throughput:** No RNC bottleneck; distributed scheduling at eNB
- **Scalability:** Stateless forwarding nodes; CP scales independently
- **Service agility:** Dynamic QoS via dedicated bearers and PCC rules
- **Cost efficiency:** IP transport throughout; commodity hardware

9.2 EPC Architecture Overview

Complete Architecture Diagram

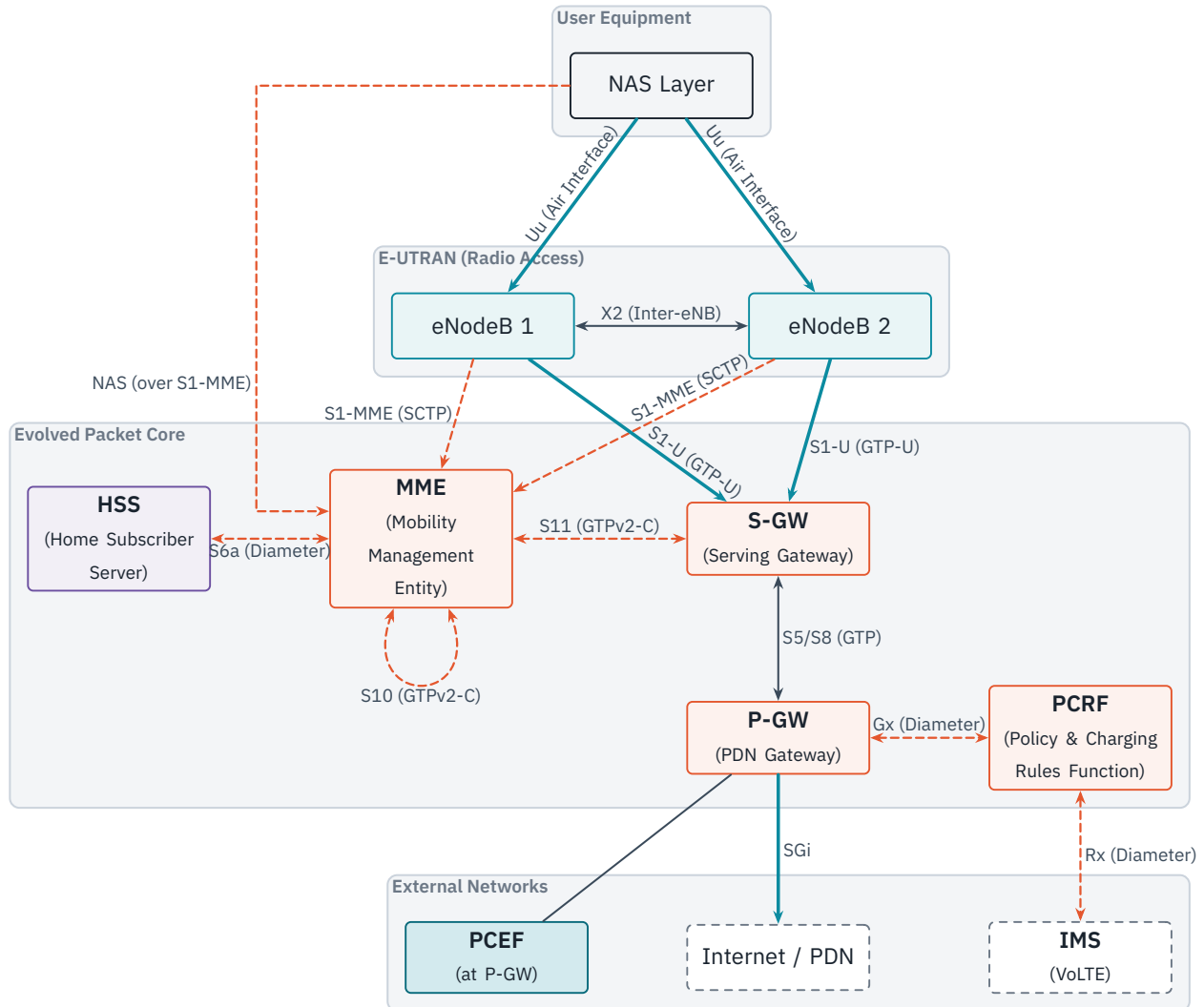


Figure 9.1 Complete EPC architecture with its interfaces and protocols

Interface Summary

Interface	Endpoints	Protocol	Purpose
S1-MME	eNB ↔ MME	S1AP over SCTP	Control plane signaling
S1-U	eNB ↔ S-GW	GTP-U over UDP	User plane data
X2	eNB ↔ eNB	X2AP + GTP-U	Handover, load mgmt
S11	MME ↔ S-GW	GTPv2-C	Session/bearer mgmt
S5/S8	S-GW ↔ P-GW	GTP (S5) or PMIP (S8)	User plane + CP
S6a	MME ↔ HSS	Diameter	Authentication, subscription
S10	MME ↔ MME	GTPv2-C	MME relocation
S3	MME ↔ SGSN	GTPv2-C	Inter-RAT mobility
Gx	PCRF ↔ P-GW	Diameter	Policy & charging rules
Gy	P-GW ↔ OCS	Diameter	Online charging
Rx	PCRF ↔ AF/IMS	Diameter	Service-level QoS request
SGi	P-GW ↔ PDN	IP	External data network

9.3 EPC Network Functions

9.3.1 MME – Mobility Management Entity

■ Purpose

The MME is the **primary control plane node** in EPC. It handles all signaling between the UE and the core network, managing mobility, security, and session establishment. The MME never touches user plane data.

■ Key Functions

1. **NAS Signaling:** Terminates Non-Access Stratum protocols (EMM and ESM) with the UE
2. **Authentication & Security:** Initiates EPS-AKA, derives keys ($K_{ASME} \rightarrow K_{eNB}$), selects NAS security algorithms
3. **Mobility Management:** Tracks UE location (TAI), manages ECM states, handles handover signaling
4. **Bearer Management:** Creates/modifies/deletes EPS bearers via S11 signaling to S-GW
5. **Paging:** Pages UEs in ECM-IDLE state when downlink data arrives
6. **MME Selection & Pooling:** Multiple MMEs serve a pool area; load balanced via S1AP weighting
7. **Inter-RAT Mobility:** Handles handover to/from 2G/3G via S3 interface to SGSN
8. **Lawful Intercept:** Provides signaling for LI triggers

■ Interfaces

Interface	Peer	Protocol	Purpose
S1-MME	eNB	S1AP/SCTP	RAN control plane
S11	S-GW	GTPv2-C	Bearer/session management
S6a	HSS	Diameter	Auth vectors, subscription data
S10	MME	GTPv2-C	MME relocation (handover)
S3	SGSN	GTPv2-C	Inter-RAT mobility
S13	EIR	Diameter	Equipment identity check
NAS	UE	EMM/ESM (over S1AP)	Direct UE signaling

■ Protocols

- **S1AP**: Application protocol for S1 interface (ASN.1 encoded, SCTP transport)
- **GTPv2-C**: GPRS Tunnelling Protocol v2 Control (UDP port 2123)
- **Diameter**: For S6a (authentication), S13 (EIR check)
- **NAS**: EMM (EPS Mobility Management) + ESM (EPS Session Management)

■ Failure Impact

- **Immediate**: All UEs served by the failed MME lose signaling connectivity
- **S1 connections released**: eNBs detect failure via SCTP heartbeat timeout
- **Recovery**: UEs in ECM-CONNECTED move to ECM-IDLE, then re-attach to a different MME in the pool
- **Mitigation**: MME pooling ensures surviving MMEs absorb load; UE context can be retrieved from HSS

9.3.2 HSS — Home Subscriber Server

■ Purpose

The HSS is the **central subscriber database** for EPC. It is the evolution of the HLR/AuC from 2G/3G, storing all permanent subscriber information, authentication credentials, and service profiles.

■ Key Functions

1. **Subscriber Database**: Stores IMSI, MSISDN, service subscriptions, APN configurations
2. **Authentication Vector Generation**: Generates EPS authentication vectors (RAND, AUTN, XRES, K_{ASME}) using the permanent key K and sequence number SQN
3. **Location Management**: Maintains current MME address per subscriber; handles registration/deregistration
4. **Subscription Profiles**: Stores allowed APNs, QoS profiles (subscribed AMBR, default QCI), roaming restrictions
5. **Charging Characteristics**: Per-subscriber charging config
6. **Service Authorization**: Determines which services a UE is permitted

■ Interfaces

Interface	Peer	Protocol	Purpose
S6a	MME	Diameter	Auth vectors, location update, subscription data
S6d	SGSN	Diameter	2G/3G equivalent of S6a
Cx/Dx	IMS (CSCF)	Diameter	IMS registration, routing
Sh	AS	Diameter	Service data for application servers

■ Protocols

- **Diameter:** All HSS interfaces use Diameter (RFC 6733) with specific applications:
 - S6a: 3GPP Diameter application (vendor-specific AVPs)
 - Authentication-Information-Request/Answer (AIR/AIA)
 - Update-Location-Request/Answer (ULR/ULA)
 - Cancel-Location-Request (CLR) for purging old MME

■ Failure Impact

- **New attaches fail:** No authentication vectors available for new UEs
- **Existing sessions survive:** MME caches auth vectors (typically 5 per UE)
- **Handovers impacted:** If new MME cannot fetch subscriber profile
- **Mitigation:** HSS is typically deployed as a geo-redundant mated pair with database replication

9.3.3 S-GW – Serving Gateway

■ Purpose

The S-GW is the **user plane anchor** between the radio network and the core. It acts as a local mobility anchor for inter-eNB handovers, ensuring user plane continuity while the UE moves between base stations.

■ Key Functions

1. **User Plane Forwarding:** Routes user data between eNB and P-GW via GTP-U tunnels
2. **Local Mobility Anchor:** During X2 handover, S-GW remains unchanged; only the downlink GTP-U endpoint is switched to the target eNB
3. **Downlink Data Buffering:** When UE is in ECM-IDLE, S-GW buffers incoming packets and triggers MME to page the UE
4. **Per-UE Charging:** Collects charging data (CDRs) per bearer, reports to charging system
5. **Lawful Intercept:** User plane copy point for legal interception
6. **Inter-RAT Anchor:** Anchors user plane during handover between LTE and 2G/3G (with SGSN)
7. **Transport Level Marking:** Applies DSCP marking based on bearer QCI

■ Interfaces

Interface	Peer	Protocol	Purpose
S1-U	eNB	GTP-U	User plane from RAN
S11	MME	GTPv2-C	Bearer/tunnel management
S5/S8	P-GW	GTP-U + GTPv2-C	User plane + signaling to P-GW
S4	SGSN	GTP	Inter-RAT user plane (2G/3G)
S12	UTRAN	GTP-U	Direct tunnel (3G)

■ Protocols

- **GTP-U** (UDP port 2152): Encapsulates user IP packets in GTP headers with TEID
- **GTPv2-C** (UDP port 2123): Create/Modify/Delete Session messages from MME
- **DSCP**: Outer IP header marking for transport QoS

■ Failure Impact

- **Active sessions disrupted**: Ongoing data flows interrupted
- **Path switch required**: MME detects failure, selects a new S-GW, re-establishes GTP tunnels
- **Buffered data lost**: Any DL packets buffered for IDLE UEs are lost
- **Mitigation**: S-GW typically deployed in N+1 redundancy; ICGW (Idle-mode Signaling Reduction) minimizes state

9.3.4 P-GW – PDN Gateway

■ Purpose

The P-GW is the **point of interconnection** between the EPC and external packet data networks (PDNs). It is the IP anchor point for UE sessions — the UE's IP address remains constant as long as the PDN connection exists, regardless of mobility.

■ Key Functions

1. IP Address Allocation:

- IPv4: DHCPv4 or static assignment from configured pools
- IPv6: Stateless Address Autoconfiguration (SLAAC) via Router Advertisement; assigns /64 prefix
- Dual-stack: Both simultaneously

2. Policy Enforcement (PCEF):

- Applies PCC rules received from PCRF
- Traffic Flow Templates (TFT): Packet filters mapping IP flows to bearers
- Service Data Flow (SDF) filters: 5-tuple matching for QoS/charging

3. Charging:

- Offline (Gz interface → CDR generation)

- Online (Gy interface → real-time credit control with OCS)
- 4. **SGi Interface:** Connects to external networks (Internet, IMS, enterprise VPN)
- 5. **NAT/NAPT:** Optional Network Address Translation for IPv4
- 6. **GTP Anchoring:** Terminates S5/S8 GTP tunnels from S-GW

■ Interfaces

Interface	Peer	Protocol	Purpose
S5/S8	S-GW	GTP	User plane + control plane
Gx	PCRF	Diameter	PCC rule download
Gy	OCS	Diameter (Ro)	Online charging
Gz	OFCS	GTP' (CDRs)	Offline charging
SGi	PDN/Internet	IP	External connectivity
S2a/S2b	ePDG/TWAG	GTP/PMIP	Wi-Fi offload

■ Protocols

- **GTP-U/GTPv2-C:** Tunnel and session management toward S-GW
- **Diameter Gx:** Policy and Charging Control (Install/Remove PCC rules)
- **Diameter Gy (Ro):** Credit-Control-Request/Answer for online charging
- **DHCPv4/ICMPv6:** IP address allocation to UE
- **RADIUS/Diameter:** Possible AAA for enterprise APNs

■ Failure Impact

- **Session loss:** All PDN connections through the failed P-GW are lost (IP addresses released)
- **UE must re-establish:** New PDN connectivity request → new IP address (breaks ongoing TCP sessions)
- **Critical impact for VoLTE:** Active voice calls dropped
- **Mitigation:** P-GW clustering, session replication, ICSR (Inter-Chassis Session Recovery)

9.3.5 PCRF – Policy and Charging Rules Function

■ Purpose

The PCRF is the **policy decision point** in the PCC architecture. It makes real-time decisions about QoS authorization, gating, and charging rules based on subscriber profile, network conditions, and application requirements.

■ Key Functions

1. **PCC Rule Generation:** Creates Policy and Charging Control rules containing:
 - SDF filters (which traffic)
 - QoS parameters (QCI, GBR, MBR)

- Charging keys and methods
- Gate status (open/close)
- 2. **QoS Authorization:** Validates requested QoS against subscription and network policy
- 3. **Dynamic Policy:** Real-time policy changes (e.g., when VoLTE call starts → install GBR bearer rule)
- 4. **Rx Interface Processing:** Receives session information from AF/IMS (media description, bandwidth) and translates to PCC rules
- 5. **Spending Limits:** Interfaces with OCS for policy decisions based on remaining credit
- 6. **Subscription-Based Policy:** Applies rules based on subscriber tier (e.g., gold/silver/bronze plans)

■ Interfaces

Interface	Peer	Protocol	Purpose
Gx	P-GW (PCEF)	Diameter	Install/remove PCC rules
Rx	AF / IMS (P-CSCF)	Diameter	Service-level info (SDP)
Sp	SPR	Diameter/proprietary	Subscriber profile retrieval
Sy	OCS	Diameter	Spending limit reporting
Sd	TDF	Diameter	Traffic Detection Function rules

■ Protocols

- **Diameter Gx Application:**
 - CC-Request/Answer (CCR/CCA): Initial, Update, Terminate
 - Re-Auth-Request (RAR): Push rule changes to PCEF
- **Diameter Rx Application:**
 - AA-Request/Answer (AAR/AAA): AF session establishment
 - Session-Termination-Request (STR): AF session teardown
 - Abort-Session-Request (ASR): Network-initiated removal

■ Failure Impact

- **Static rules continue:** Previously installed PCC rules at P-GW remain active
- **No dynamic policy:** New service requests (e.g., VoLTE) cannot get dedicated bearer authorization
- **Default QoS only:** New sessions get only default bearer with static policy
- **Mitigation:** PCRF deployed in active/standby pairs; Diameter routing agent (DRA) for failover

9.3.6 PCEF – Policy and Charging Enforcement Function

■ Purpose

The PCEF is the **policy enforcement point**, co-located with the P-GW. It enforces the PCC rules received from the PCRF by applying packet filtering, QoS enforcement, and charging actions on every user plane packet.

■ Key Functions

1. **SDF Filter Enforcement:** Matches incoming/outgoing IP packets against Service Data Flow filters (5-tuple: src IP, dst IP, src port, dst port, protocol)
2. **Gate Control:** Opens or closes the gate for specific SDFs (blocking/allowing traffic)
3. **QoS Enforcement:** Applies rate limiting per-flow:
 - GBR (Guaranteed Bit Rate): Minimum rate ensured
 - MBR (Maximum Bit Rate): Hard cap per bearer
 - AMBR (Aggregate Maximum Bit Rate): Cap across all non-GBR bearers
4. **Bearer Binding:** Maps SDF filters to appropriate EPS bearers
5. **Charging Enforcement:**
 - Applies charging keys to traffic
 - Volume/time measurement per rating group
 - Online charging: Reports usage, enforces credit limits
 - Offline charging: Generates CDRs
6. **Usage Monitoring:** Reports volume thresholds to PCRF for policy decisions

■ Interfaces

The PCEF is integrated within the P-GW, so its external interface is primarily:

Interface	Peer	Protocol	Purpose
Gx	PCRF	Diameter	Receives PCC rules
Gy	OCS	Diameter (Ro)	Online charging credit control
Gz	OFCS	GTP'	Offline CDR delivery

■ Protocols

- **Diameter Gx:** Receives Charging-Rule-Install/Remove AVPs containing PCC rules
- **Deep Packet Inspection (DPI):** Optional; identifies applications for application-aware charging
- **Token Bucket / Leaky Bucket:** Rate enforcement algorithms for GBR/MBR policing

■ Failure Impact

- Since PCEF is co-located with P-GW, PCEF failure = P-GW failure (see Section 9.3.4)
- **Degraded mode:** If only policy engine fails, traffic may pass unpoliced (fail-open) or be blocked (fail-close) depending on implementation

- **Mitigation:** Hardware redundancy within P-GW chassis; hitless software upgrades

9.4 EPC Bearers

■ Bearer Concept

An EPS bearer is an **end-to-end logical channel** between the UE and the P-GW with specific QoS characteristics. Every bearer is defined by three components:

$$\text{Bearer} = \{ \text{TFT} + \text{QoS parameters} + \text{GTP tunnel(s)} \}$$

- **TFT (Traffic Flow Template):** Packet filters (5-tuple) that classify which IP packets belong to this bearer
- **QoS Parameters:** QCI, ARP, GBR/MBR (for GBR bearers), AMBR (aggregate)
- **GTP Tunnel:** Identified by TEID (Tunnel Endpoint Identifier) on each hop:
 - S1-U: eNB ↔ S-GW
 - S5/S8: S-GW ↔ P-GW

■ Bearer Structure Diagram

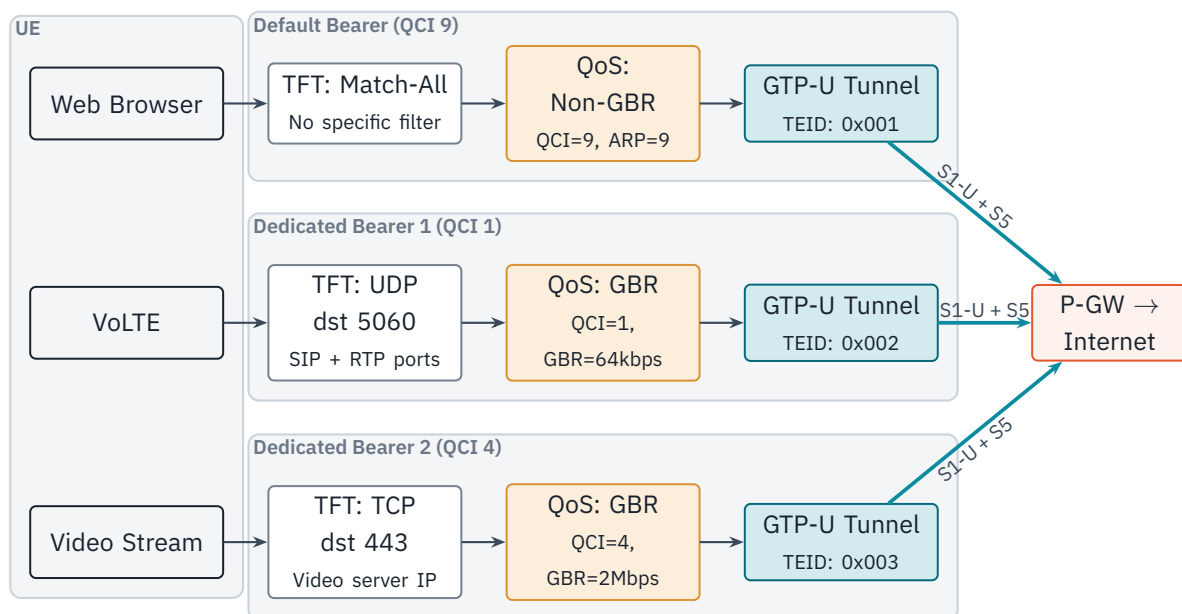


Figure 9.2 Default and dedicated bearers: TFT, QoS and GTP-U tunnel per bearer

9.4.1 Default Bearer

- **Created automatically** during PDN connection (Attach or additional PDN connectivity request)
- **Always-on:** Exists for the entire duration of the PDN connection
- **Non-GBR:** No guaranteed bit rate; best-effort with AMBR cap

- **Match-all TFT:** Carries all traffic that doesn't match any dedicated bearer's TFT
- **QoS from subscription:** QCI and AMBR values come from HSS subscription profile
- **One per PDN connection:** Each APN gets exactly one default bearer (EBI = EPS Bearer ID assigned)

9.4.2 Dedicated Bearer

- **Created on-demand:** Triggered by service request (PCRF rule) or UE/network request
- **Specific QoS:** Can be GBR (guaranteed) or Non-GBR (prioritized best-effort)
- **Specific TFT:** Packet filters direct matching traffic to this bearer
- **Linked to default bearer:** Shares the same PDN connection (same IP address)
- **Lifecycle:** Created via Bearer Resource Allocation → PCRF decision → P-GW initiates dedicated bearer activation
- **Example triggers:** VoLTE call start (QCI 1), video streaming (QCI 4), gaming (QCI 3)

9.4.3 QCI Table (Standardized Values)

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	GBR	2	100 ms	10^{-2}	Conversational Voice (VoLTE)
2	GBR	4	150 ms	10^{-3}	Conversational Video (live)
3	GBR	3	50 ms	10^{-3}	Real-time Gaming
4	GBR	5	300 ms	10^{-6}	Non-conversational Video (buffered streaming)
5	Non-GBR	1	100 ms	10^{-6}	IMS Signaling (SIP)
6	Non-GBR	6	300 ms	10^{-6}	Video (buffered), TCP-based services (web, email)
7	Non-GBR	7	100 ms	10^{-3}	Voice, Video (live), Interactive Gaming
8	Non-GBR	8	300 ms	10^{-6}	Video (buffered), TCP-based (premium)
9	Non-GBR	9	300 ms	10^{-6}	Video (buffered), TCP-based (default/best-effort)

Key QoS Parameters:

- **QCI (QoS Class Identifier):** Scalar value (1-9) mapping to standardized packet treatment

- **ARP (Allocation and Retention Priority):** Priority level (1-15), pre-emption capability/vulnerability
- **GBR:** Guaranteed minimum bit rate (only for QCI 1-4)
- **MBR:** Maximum bit rate (caps GBR bearers)
- **UE-AMBR:** Aggregate MBR across all non-GBR bearers of a UE
- **APN-AMBR:** Aggregate MBR across all non-GBR bearers of one APN

9.5 Authentication in EPC

9.5.1 EPS-AKA (EPS Authentication and Key Agreement)

EPS-AKA provides **mutual authentication** between the UE and the network. Both parties prove knowledge of the permanent key K (stored in USIM and HSS) without transmitting it.

■ EPS-AKA Flow

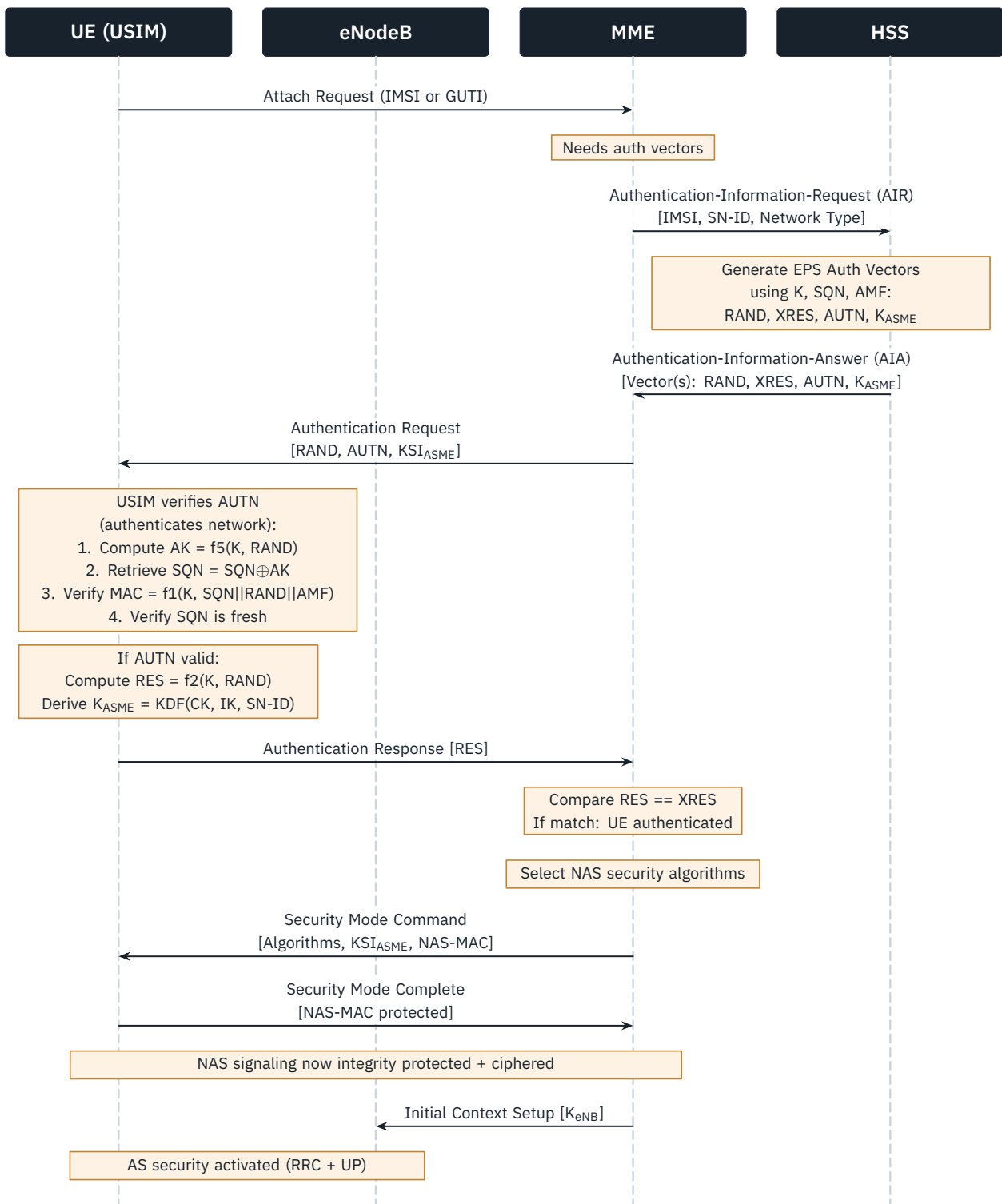


Figure 9.3 EPS-AKA authentication and NAS/AS security activation

■ Authentication Failure Cases

- **MAC failure** (UE rejects AUTN): Network is not authentic → UE sends Authentication Failure (cause: MAC failure)
- **SQN out of range**: Synchronization failure → UE sends AUTS for resync
- **XRES mismatch**: UE is not authentic → MME sends Authentication Reject

9.5.2 K_{ASME} Derivation

The K_{ASME} (Key for Access Security Management Entity) is the master key for all EPC security:

$$K_{ASME} = \text{KDF}(\text{CK} \parallel \text{IK}, \text{SN-ID}, \text{SQN} \oplus \text{AK})$$

where:

- KDF = HMAC-SHA-256 based key derivation function
- CK = cipher key = $f_3(K, \text{RAND})$
- IK = integrity key = $f_4(K, \text{RAND})$
- SN-ID = serving network identity (MCC || MNC)
- SQN = sequence number
- AK = anonymity key = $f_5(K, \text{RAND})$

■ Key Hierarchy

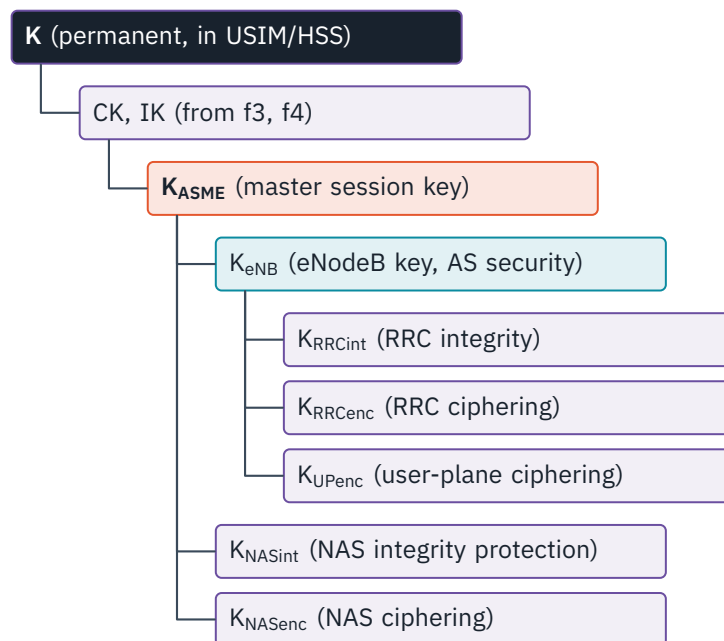


Figure 9.4 EPS key hierarchy

9.5.3 NAS Security

NAS security operates between UE and MME (transparent to eNB):

Protection	Algorithm Options	Purpose
Integrity (mandatory)	EIA1 (SNOW 3G), EIA2 (AES), EIA3 (ZUC)	Prevents message tampering
Ciphering (optional)	EEA0 (null), EEA1 (SNOW 3G), EEA2 (AES), EEA3 (ZUC)	Prevents eavesdropping

- **NAS COUNT:** 32-bit counter prevents replay attacks (separate UL/DL counts)
- **Security Mode Command:** Establishes algorithms; itself integrity-protected
- **Protected messages:** All NAS messages after SMC are integrity-protected; most are ciphered

9.5.4 AS Security (Access Stratum)

AS security operates between UE and eNB:

- K_{eNB} : Derived from K_{ASME} at MME, sent to eNB in Initial Context Setup
- **RRC protection:** Both integrity and ciphering (SRB1, SRB2)
- **User plane:** Ciphering only (no integrity for performance reasons in LTE)
- **Key refresh:** New K_{eNB} derived at handover ($K_{eNB}^* = \text{KDF}(K_{eNB}, \text{PCI}, \text{freq})$)
- **Activated via:** SecurityModeCommand (RRC level, distinct from NAS SMC)

9.6 Session Management

9.6.1 PDN Connectivity (Default Bearer Establishment)

PDN connectivity creates the **default EPS bearer** and assigns an IP address to the UE.

■ Procedure Flow

1. **UE → MME:** ESM PDN Connectivity Request (APN, PDN type: IPv4/IPv6/IPv4v6)
2. **MME → S-GW:** Create Session Request (IMSI, bearer QoS from HSS profile, APN)
3. **S-GW → P-GW:** Create Session Request (forwarded with S-GW F-TEID)
4. **P-GW:** Allocates IP address, contacts PCRF (CCR-Initial on Gx), installs default PCC rules
5. **P-GW → S-GW:** Create Session Response (IP address, bearer context, TFT)
6. **S-GW → MME:** Create Session Response (S-GW F-TEID, bearer context)
7. **MME → eNB:** Initial Context Setup Request (bearer context, K_{eNB} , NAS: Attach Accept + Activate Default Bearer)
8. **eNB → UE:** RRC Connection Reconfiguration (DRB setup) + NAS messages
9. **UE → eNB → MME:** Attach Complete / Activate Default EPS Bearer Context Accept
10. **MME → S-GW:** Modify Bearer Request (eNB F-TEID for downlink)

Result: End-to-end bearer established: UE ↔ eNB ↔ S-GW ↔ P-GW ↔ Internet

9.6.2 Dedicated Bearer Activation

Dedicated bearers are **network-initiated** (triggered by PCRF rule or operator policy):

1. **PCRF** → **P-GW**: Re-Auth-Request (RAR) with PCC rule (QCI, GBR, TFT filters)
2. **P-GW** → **S-GW**: Create Bearer Request (EBI, TFT, QoS, S5 F-TEID)
3. **S-GW** → **MME**: Create Bearer Request (forwarded with S1-U F-TEID)
4. **MME** → **eNB** → **UE**: Bearer Setup Request (NAS: Activate Dedicated Bearer Context Request)
5. **UE**: Accepts, stores TFT for uplink classification
6. **UE** → **eNB** → **MME**: Activate Dedicated EPS Bearer Context Accept
7. **MME** → **S-GW** → **P-GW**: Create Bearer Response (all TEIDs now bound)

9.6.3 Bearer Modification

Bearer modification changes QoS or TFT of an existing bearer:

- **Network-initiated**: PCRF updates PCC rule → P-GW sends Update Bearer Request
- **UE-initiated**: UE sends Bearer Resource Modification Request to MME
- **Examples**: Upgrade video quality (increase GBR), add new SDF filter, change gate status

9.6.4 Bearer Deactivation

- **Network-initiated**: PCRF removes PCC rule → P-GW sends Delete Bearer Request
- **UE-initiated**: UE sends Deactivate EPS Bearer Context Request
- **Default bearer deactivation**: Deletes ALL associated dedicated bearers + releases IP address
- **PDN disconnection**: Triggered when last default bearer is deactivated

9.7 Mobility Management

9.7.1 ECM States

State	Description	UE Location Known?	Resources
ECM-IDLE	No NAS signaling connection; UE monitors paging	TA-level only (TAI list)	No S1 connection, no DRB; S-GW buffers DL data
ECM-CONNECTED	Active NAS signaling via S1-MME	Cell-level (ECGI)	S1-U tunnel active; DRBs established

State Transitions:

- **IDLE** → **CONNECTED**: Service Request (UE-triggered) or Paging Response (network-triggered)
- **CONNECTED** → **IDLE**: S1 Release (inactivity timer expiry or explicit release)

9.7.2 Tracking Area Update (TAU)

TAU keeps the network informed of the UE's approximate location in ECM-IDLE state:

When TAU is triggered:

- UE enters a new Tracking Area (TA) not in its TAI list
- Periodic TAU timer expires (configurable, typically 54 minutes)
- After ISR (Idle-mode Signaling Reduction) activation

TAU Procedure:

1. UE → MME: Tracking Area Update Request (old GUTI, TAI visited)
2. MME: Verifies identity, optionally authenticates
3. MME → HSS: Update Location (if new MME)
4. MME → old MME: Context transfer (if MME changed)
5. MME → UE: TAU Accept (new GUTI, new TAI list)
6. MME → S-GW: Modify Bearer (if S-GW changed)

TAI List: MME assigns a list of TAs to reduce TAU frequency (load balancing vs. paging cost tradeoff)

9.7.3 X2 Handover (Intra-LTE, Without MME Change)

The **fastest handover** in LTE — direct eNB-to-eNB signaling without MME involvement for data forwarding:

1. **Source eNB:** Measurement reports indicate target cell is better
2. **Source → Target eNB (X2):** Handover Request (UE context, bearers, QoS)
3. **Target eNB:** Admits UE, allocates resources
4. **Target → Source eNB (X2):** Handover Request Acknowledge (DL forwarding address)
5. **Source eNB → UE:** RRC Connection Reconfiguration (handover command)
6. **Source eNB:** Starts forwarding buffered + new DL data to target eNB via X2
7. **UE:** Detaches from source, synchronizes to target cell
8. **UE → Target eNB:** RRC Connection Reconfiguration Complete
9. **Target eNB → MME:** Path Switch Request (new ECGI, new TEIDs)
10. **MME → S-GW:** Modify Bearer Request (target eNB address)
11. **S-GW:** Switches DL path to target eNB (end marker to source)
12. **MME → Target eNB:** Path Switch Request Acknowledge
13. **Target → Source eNB:** UE Context Release (cleanup)

Key point: S-GW is unchanged (local mobility anchor). Only the S1-U endpoint moves.

9.7.4 S1 Handover (With MME/S-GW Change)

Used when X2 is unavailable or when MME/S-GW relocation is needed:

1. **Source eNB → Source MME:** Handover Required (target ID, cause)
2. **Source MME → Target MME:** Forward Relocation Request (UE context) [if MME changes]
3. **Target MME → Target eNB:** Handover Request (bearer setup)

4. **Target eNB** → **Target MME**: Handover Request Acknowledge
5. **Target MME** → **Source MME**: Forward Relocation Response
6. **Source MME** → **Source eNB**: Handover Command
7. **Source eNB** → **UE**: RRC Connection Reconfiguration
8. **UE** → **Target eNB**: Handover Confirm
9. **Target eNB** → **Target MME**: Handover Notify
10. **Target MME** → **S-GW**: Modify Bearer / Create Session (if S-GW changes)

Indirect Data Forwarding: DL data forwarded Source eNB → Source S-GW → Target S-GW → Target eNB during handover gap.

9.7.5 Inter-RAT Mobility (LTE ↔ 3G)

■ LTE → 3G (PS Handover)

1. MME triggers handover to UTRAN (measurement reports or coverage)
2. MME → SGSN (S3): Forward Relocation Request (PDP contexts mapped from EPS bearers)
3. SGSN → RNC: Relocation Request
4. After handover: S-GW reroutes user plane via S4 to SGSN (or direct tunnel via S12 to RNC)
5. Bearer QoS mapped: QCI → UMTS Traffic Class (Conversational/Streaming/Interactive/Background)

■ 3G → LTE

1. SGSN detects LTE capability; initiates handover
2. SGSN → MME (S3): Forward Relocation Request
3. MME activates bearers, sets up S1 path
4. PDP contexts mapped to EPS bearers

9.8 Failure Scenarios

9.8.1 MME Failure

Aspect	Impact
Detection	eNB detects via SCTP association failure (heartbeat timeout ~30s)
Immediate effect	All S1-MME connections to failed MME are lost
ECM-CONNECTED UEs	Move to ECM-IDLE; eNB releases radio resources
Recovery action	UE sends Service Request / Attach to new MME (selected via S1AP weighting from MME pool)
Context recovery	New MME fetches subscription from HSS; active bearers re-established via new Create Session to S-GW
Data loss	Momentary — ongoing TCP sessions may survive (IP address preserved at P-GW); UDP/real-time sessions interrupted

Mitigation Strategies:

- MME pooling (multiple MMEs per tracking area)
- Active/Standby with session state replication
- S10-based context transfer from backup node
- DNS-based MME selection for load distribution

9.8.2 S-GW Failure

Aspect	Impact
Detection	MME detects via GTPv2-C Echo timeout on S11; eNB detects via GTP-U path failure on S1-U
Immediate effect	User plane interrupted for all UEs served by failed S-GW
Recovery action	MME selects new S-GW, sends Modify Bearer to P-GW with new S-GW address
Path switch	P-GW redirects S5 tunnel to new S-GW; MME updates eNB with new S1-U TEID
Data loss	Buffered DL data (for IDLE UEs) at failed S-GW is lost

Mitigation Strategies:

- S-GW pool with DNS-based selection (weighted records)
- GTP-U path failure detection (Echo Request/Response)
- N+1 redundancy; ICSR between chassis pairs

- Pre-established backup GTP contexts

9.8.3 P-GW Failure

Aspect	Impact
Detection	S-GW detects via GTP path failure on S5; PCRF detects via Diameter transport failure on Gx
Immediate effect	All PDN connections through failed P-GW are lost
IP address loss	UE IP addresses are released — cannot be recovered
Session impact	All TCP/UDP sessions terminated; VoLTE calls dropped
Recovery action	UE must establish new PDN connectivity (new default bearer, new IP address)
Application impact	DNS-based services recover; long-lived sessions (VPN, SSH) broken

Mitigation Strategies:

- Inter-Chassis Session Recovery (ICSR): Active/Standby P-GW pair with session state sync
- GTP-C restart counter: S-GW detects P-GW restart, reports to MME
- IP address pool replication between paired P-GWs
- P-GW selection via DNS with health monitoring

9.8.4 HSS Unreachable

Aspect	Impact
Detection	MME detects via Diameter transport failure or watchdog timeout on S6a
Existing sessions	Continue normally — bearer state is at MME/S-GW/P-GW
Authentication	MME uses cached authentication vectors (typically 5 vectors per UE fetched at last AIR)
New attaches	Fail once cached vectors exhausted — no new UEs can authenticate
Handover to new MME	Impacted — new MME cannot fetch subscription profile
Duration tolerance	Depends on cache depth; typically sustains operations for minutes to hours

Mitigation Strategies:

- Geo-redundant HSS pair (active/active or active/standby)
- Diameter Routing Agent (DRA) for automatic failover
- Increased auth vector batch size (e.g., 10 instead of 5)
- MME-level subscription caching with TTL

9.9 Summary

Component	Plane	Primary Role	Key Interface
MME	Control	Signaling, Auth, Mobility	S1-MME, S11, S6a
HSS	Control	Subscriber DB, Auth vectors	S6a
S-GW	User + Control	Local UP anchor, buffering	S1-U, S5/S8
P-GW	User + Control	IP anchor, policy enforcement	S5/S8, SGi, Gx
PCRF	Control	Policy decision	Gx, Rx
PCEF	User	Policy enforcement	Gx (at P-GW)

Key Takeaways

1. **EPC eliminates circuit-switching:** All services (including voice) over IP bearers
2. **Control/User plane split:** Enables independent scaling and evolution toward 5G
3. **Bearer-based QoS:** Deterministic quality for real-time services (QCI 1-4 = GBR)
4. **Security by design:** Mutual authentication (EPS-AKA), layered key derivation, mandatory integrity
5. **Mobility without disruption:** GTP anchoring at S-GW (local) and P-GW (global) preserves IP sessions
6. **Policy architecture:** PCRF/PCEF enable real-time, subscriber-aware traffic treatment
7. **Resilience through pooling:** MME pools, S-GW pools, and HSS redundancy minimize single points of failure

Module 9 – End